

Android Lab2: 菜单 UI 设计报告

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菜单 UI 设计-version1.0 (运行于 Android Studio 2020.3.1)

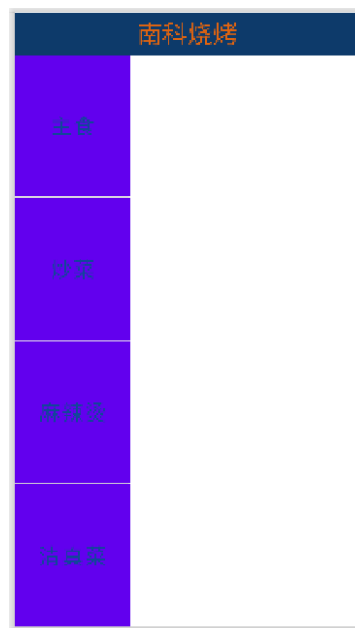


图 1: 菜单 version1.0 UI 界面

实现初步的布局及功能:

- 1、实现初步的布局。

Version1.0 版本存在的问题:

- 1、颜色变化。
- 2、Item 仍未编程。
- 3、代码理解难度大。

优化过程



图 2：Android Studio 2024.3.1 显示界面

优化 1：改用最新版的 Android Studio，方便对代码进行 debug。

How to Fix the "Namespace not specified" Error in Android Gradle

This error occurs because newer versions of Android Gradle Plugin (AGP) require you to specify a namespace in your module's `build.gradle` file rather than relying on the package name in the `AndroidManifest.xml`.

图 3：Nexus 6 API30 显示界面

优化 2：最新版的 Android Studio 需要添加代码

```
1 namespace 'com.example.week7'
```

菜单 UI 设计-version2.0（运行于 Android Studio 2024.3.1）

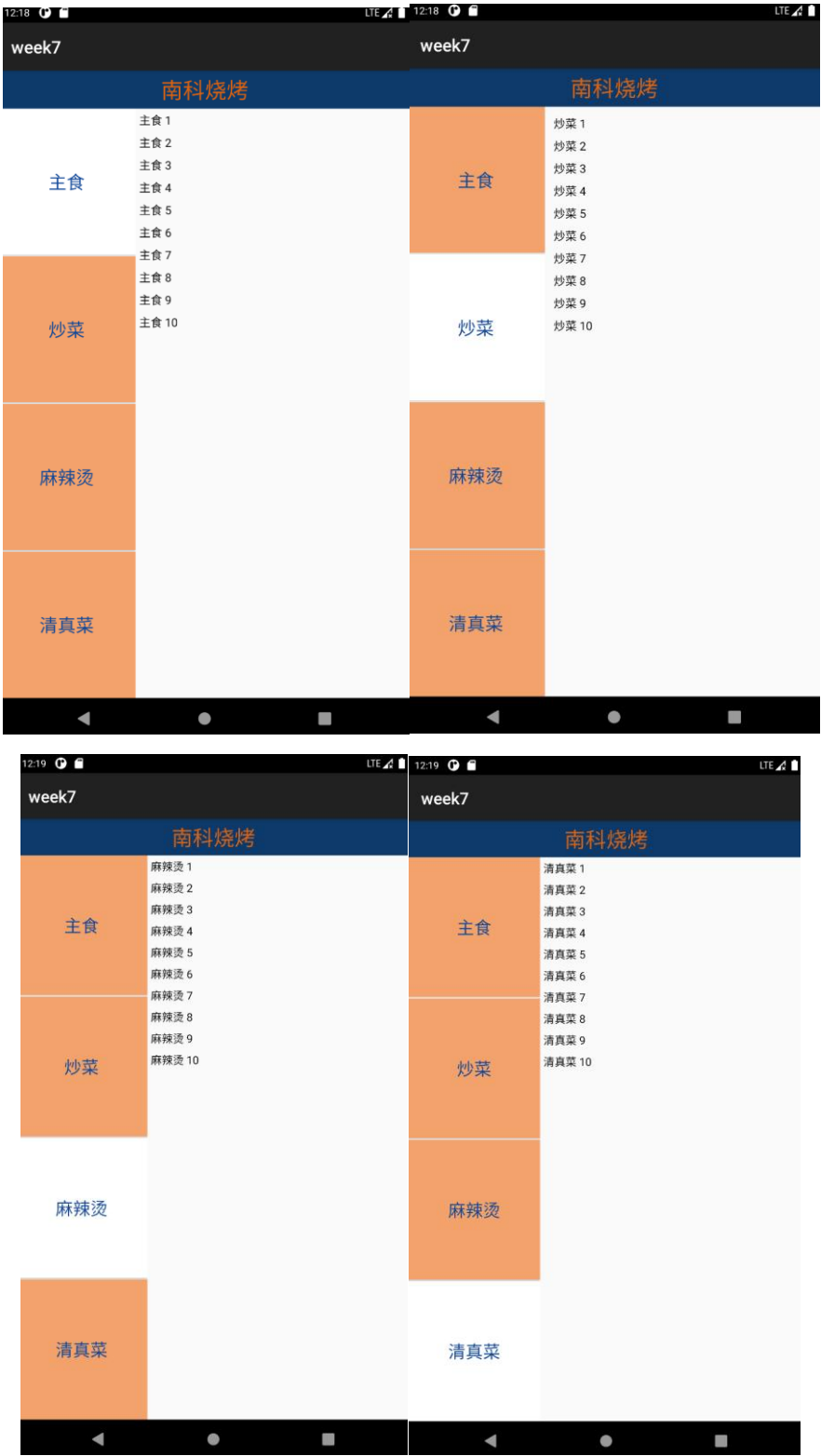


图 4：菜单 UI（最终版）

实现布局及功能：

- 1、选中的按钮呈现白色。
- 2、除选中的按钮外，其他按钮呈现橙色。
- 3、每个选项包含十项菜品。
- 4、实现基础的排版。
- 5、进一步美化。

附：项目完整代码

activity_main.xml

```
01 <?xml version="1.0" encoding="utf-8"?>
02 <LinearLayout
03     xmlns:android="http://schemas.android.com/apk/res/android"
04     xmlns:app="http://schemas.android.com/apk/res-auto"
05     xmlns:tools="http://schemas.android.com/tools"
06     android:layout_width="match_parent"
07     android:layout_height="match_parent"
08     tools:context=".MainActivity"
09     android:orientation="vertical">
10     <TextView
11         android:layout_width="match_parent"
12         android:layout_height="wrap_content"
13         android:text="南科烧烤"
14         android:gravity="center"
15         android:textColor="#DF6411"
16         android:textSize="30sp"
17         android:background="#0D3A6A"
18         android:padding="5dp"/>
19     <LinearLayout
20         android:id="@+id/ll_menu"
21         android:layout_width="match_parent"
22         android:layout_height="match_parent"
23         android:orientation="horizontal">
24         <LinearLayout
25             android:layout_width="0dp"
26             android:layout_height="match_parent"
27             android:layout_weight="1"
28             android:orientation="vertical">
```

```
29         <Button
30             android:id="@+id/btnzhushi"
31             android:layout_width="match_parent"
32             android:layout_height="0dp"
33             android:text="主食"
34             android:textSize="25sp"
35             android:textColor="#104798"
36             android:background="#F3A26C"
37             android:layout_weight="1"/>
38         <Button
39             android:id="@+id/btnchaocai"
40             android:layout_width="match_parent"
41             android:layout_height="0dp"
42             android:text="炒菜"
43             android:textSize="25sp"
44             android:textColor="#104798"
45             android:background="#F3A26C"
46             android:layout_weight="1"
47             android:layout_marginTop="2dp"/>
48         <Button
49             android:id="@+id/btnmalat"
50             android:layout_width="match_parent"
51             android:layout_height="0dp"
52             android:text="麻辣烫"
53             android:textSize="25sp"
54             android:textColor="#104798"
55             android:background="#F3A26C"
56             android:layout_weight="1"
57             android:layout_marginTop="2dp"/>
58         <Button
59             android:id="@+id/btnqingzhencai"
60             android:layout_width="match_parent"
61             android:layout_height="0dp"
62             android:text="清真菜"
63             android:textSize="25sp"
64             android:textColor="#104798"
65             android:background="#F3A26C"
66             android:layout_weight="1"
67             android:layout_marginTop="2dp"/>
68     </LinearLayout>
69     <LinearLayout
70         android:id="@+id/ll_detail"
71         android:layout_width="0dp"
72         android:layout_height="match_parent"
```

```

73         android:layout_weight="2"
74         android:orientation="horizontal">
75     </LinearLayout>
76 </LinearLayout>
77
78
</LinearLayout>

```

fragment_chaocai.xml

```

01 <?xml version="1.0" encoding="utf-8"?>
02
03 <LinearLayout
04     xmlns:android="http://schemas.android.com/apk/res/android"
05     android:layout_width="match_parent"
06     android:layout_height="match_parent"
07     android:orientation="vertical">
08
09     <androidx.recyclerview.widget.RecyclerView
10         android:id="@+id/recycler_view"
11         android:layout_width="match_parent"
12         android:layout_height="match_parent"
13         android:padding="6dp" />
14
</LinearLayout>

```

fragment_malat.xml

```

01 <?xml version="1.0" encoding="utf-8"?>
02 <androidx.constraintlayout.widget.ConstraintLayout
03     xmlns:android="http://schemas.android.com/apk/res/android"
04     xmlns:app="http://schemas.android.com/apk/res-auto"
05     xmlns:tools="http://schemas.android.com/tools"
06     android:layout_width="match_parent"
07     android:layout_height="match_parent"
08     tools:context=".malatFragment">
09
10     <androidx.recyclerview.widget.RecyclerView
11         android:id="@+id/recycler_view"
12         android:layout_width="match_parent"
13         android:layout_height="match_parent"
14         app:layout_constraintTop_toTopOf="parent"

```

```

15         app:layout_constraintBottom_toBottomOf="parent"
16         app:layout_constraintStart_toStartOf="parent"
17         app:layout_constraintEnd_toEndOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

fragment_qingzhencai.xml

```

01 <?xml version="1.0" encoding="utf-8"?>
02 <androidx.constraintlayout.widget.ConstraintLayout
03     xmlns:android="http://schemas.android.com/apk/res/android"
04     xmlns:app="http://schemas.android.com/apk/res-auto"
05     xmlns:tools="http://schemas.android.com/tools"
06     android:layout_width="match_parent"
07     android:layout_height="match_parent"
08     tools:context=".qingzhencaiFragment">
09
10     <androidx.recyclerview.widget.RecyclerView
11         android:id="@+id/recycler_view"
12         android:layout_width="match_parent"
13         android:layout_height="match_parent"
14         app:layout_constraintTop_toTopOf="parent"
15         app:layout_constraintBottom_toBottomOf="parent"
16         app:layout_constraintStart_toStartOf="parent"
17         app:layout_constraintEnd_toEndOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

fragment_zhushi.xml

```

01 <?xml version="1.0" encoding="utf-8"?>
02 <androidx.constraintlayout.widget.ConstraintLayout
03     xmlns:android="http://schemas.android.com/apk/res/android"
04     xmlns:app="http://schemas.android.com/apk/res-auto"
05     xmlns:tools="http://schemas.android.com/tools"
06     android:layout_width="match_parent"
07     android:layout_height="match_parent"
08     tools:context=".zhushiFragment">
09
10     <androidx.recyclerview.widget.RecyclerView

```

```

11         android:id="@+id/recycler_view"
12         android:layout_width="match_parent"
13         android:layout_height="match_parent"
14         app:layout_constraintTop_toTopOf="parent"
15         app:layout_constraintBottom_toBottomOf="parent"
16         app:layout_constraintStart_toStartOf="parent"
17         app:layout_constraintEnd_toEndOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

itemlayout.xml

```

01 <?xml version="1.0" encoding="utf-8"?>
02 <LinearLayout
03     xmlns:android="http://schemas.android.com/apk/res/android"
04     android:layout_width="match_parent"
05     android:layout_height="wrap_content"
06     android:orientation="vertical">
07
08     <TextView
09         android:id="@+id/item_text"
10         android:layout_width="wrap_content"
11         android:layout_height="wrap_content"
12         android:textColor="#000000"
13         android:textSize="16sp"
14         android:padding="4dp" />
</LinearLayout>

```


MainActivity

```
001 package com.example.week7;
002
003 import androidx.annotation.NonNull;
004 import androidx.appcompat.app.AppCompatActivity;
005 import androidx.recyclerview.widget.LinearLayoutManager;
006 import androidx.recyclerview.widget.RecyclerView;
007
008 import android.animation.ObjectAnimator;
009 import android.annotation.SuppressLint;
010 import android.content.ComponentName;
011 import android.content.Intent;
012 import android.content.ServiceConnection;
013 import android.graphics.Color;
014 import android.media.MediaPlayer;
015 import android.os.Bundle;
016 import android.os.Handler;
017 import android.os.HandlerThread;
018 import android.os.IBinder;
019 import android.os.Looper;
020 import android.os.Message;
021 import android.view.View;
022 import android.view.ViewGroup;
023 import android.view.animation.LinearInterpolator;
024 import android.widget.Button;
025 import android.widget.ImageView;
026 import android.widget.SeekBar;
027 import android.widget.TextView;
028
029 import java.util.Timer;
```

```

030
031 public class MainActivity extends AppCompatActivity{
032     private zhushiFragment zhushiFragment = new zhushiFragment();//声明 Fragment
033     private chaocaiFragment chaocaiFragment = new chaocaiFragment();
034     private malatFragment malatFragment=new malatFragment();
035     private qingzhencaiFragment qingzhencaiFragment=new qingzhencaiFragment();
036     private Button btn_zhushi,btn_chaocai,btn_mlt,btn_qzc;
037
038
039
040     @SuppressWarnings("MissingInflatedId")
041     @Override
042     protected void onCreate(Bundle savedInstanceState) {
043         super.onCreate(savedInstanceState);
044         setContentView(R.layout.activity_main);
045
046         getSupportFragmentManager().beginTransaction().add(R.id.ll_detail,zhushiFragment).commitAllowingStateLo
047 ss();//绑定 Fragment
048         btn_zhushi = findViewById(R.id.btnzhushi);
049         btn_chaocai = findViewById(R.id.btnchaocai);
050         btn_mlt= findViewById(R.id.btnmalat);
051         btn_qzc= findViewById(R.id.btnqingzhencai);
052
053         OnClickListener onClick = new OnClickListener();
054         btn_zhushi.setOnClickListener(onClick);
055         btn_chaocai.setOnClickListener(onClick);
056         btn_mlt. setOnClickListener(onClick);
057         btn_qzc.setOnClickListener(onClick);
058
059         btn_zhushi.performClick();//自动点击
060
061
062     }
063
064
065
066
067     private class OnClickListener implements View.OnClickListener{
068         @Override
069         public void onClick(View view) {
070             if(view.getId()==R.id.btnzhushi){
071                 btn_zhushi.setBackgroundColor(getResources().getColor(R.color.white));
072                 btn_chaocai.setBackgroundColor(getResources().getColor(R.color.oringin));
073                 btn_mlt.setBackgroundColor(getResources().getColor(R.color.oringin));

```

```

074         btn_qzc. setBackgroundColor(getResources().getColor(R.color.oringin));
075         getSupportFragmentManager().beginTransaction().replace(R.id.ll_detail,zhushiFragment).commitAll
076 owingStateLoss();
077     }
078     else if(view.getId()==R.id.btnchaocai){
079         btn_zhushi.setBackgroundColor(getResources().getColor(R.color.oringin));
080         btn_chaocai.setBackgroundColor(getResources().getColor(R.color.white));
081         btn_mlt.setBackgroundColor(getResources().getColor(R.color.oringin));
082         btn_qzc. setBackgroundColor(getResources().getColor(R.color.oringin));
083         getSupportFragmentManager().beginTransaction().replace(R.id.ll_detail,chaocaiFragment).commitAl
084 lowingStateLoss();
085     }
086     else if(view.getId()==R.id.btnmalat){
087         btn_zhushi.setBackgroundColor(getResources().getColor(R.color.oringin));
088         btn_chaocai.setBackgroundColor(getResources().getColor(R.color.oringin));
089         btn_mlt.setBackgroundColor(getResources().getColor(R.color.white));
090         btn_qzc. setBackgroundColor(getResources().getColor(R.color.oringin));
091         getSupportFragmentManager().beginTransaction().replace(R.id.ll_detail,malatFragment).commitAllo
092 wingStateLoss();
093     }
094     }
095     else if(view.getId()==R.id.btnqingzhencai){
096         btn_zhushi.setBackgroundColor(getResources().getColor(R.color.oringin));
097         btn_chaocai.setBackgroundColor(getResources().getColor(R.color.oringin));
098         btn_mlt.setBackgroundColor(getResources().getColor(R.color.oringin));
099         btn_qzc. setBackgroundColor(getResources().getColor(R.color.white));
100         getSupportFragmentManager().beginTransaction().replace(R.id.ll_detail,qingzhencaiFragment).comm
101 itAllowingStateLoss();
102     }
103     }
104     }
105 }

```

chaocaiFragment

```

01 package com.example.week7;
02
03 import androidx.fragment.app.Fragment;
04 import androidx.recyclerview.widget.LinearLayoutManager;
05 import androidx.recyclerview.widget.RecyclerView;
06
07 import android.os.Bundle;
08 import android.view.LayoutInflater;

```

```

09 import android.view.View;
10 import android.view.ViewGroup;
11
12 import java.util.ArrayList;
13 import java.util.List;
14
15 public class chaocaiFragment extends Fragment {
16
17     private RecyclerView recyclerView;
18     private ItemAdapter itemAdapter;
19
20     @Override
21     public View onCreateView(LayoutInflater inflater, ViewGroup
22 container,
23
24                                     Bundle savedInstanceState) {
25
26         View view = inflater.inflate(R.layout.fragment_chaocai,
27 container, false);
28
29         recyclerView = view.findViewById(R.id.recycler_view);
30         recyclerView.setLayoutManager(new
31 LinearLayoutManager(getContext()));
32
33         List<String> itemList = new ArrayList<>();
34         for (int i = 1; i <= 10; i++) {
35             itemList.add("炒菜 " + i );
36         }
37
38         itemAdapter = new ItemAdapter(getContext(), itemList);
39         recyclerView.setAdapter(itemAdapter);
40
41         return view;
42     }
43 }

```

malatFragment

```

01 package com.example.week7;
02 import androidx.fragment.app.Fragment;
03 import androidx.recyclerview.widget.LinearLayoutManager;
04 import androidx.recyclerview.widget.RecyclerView;
05
06 import android.os.Bundle;

```

```

07 import android.view.LayoutInflater;
08 import android.view.View;
09 import android.view.ViewGroup;
10
11 import java.util.ArrayList;
12 import java.util.List;
13
14 public class malatFragment extends Fragment {
15
16     private RecyclerView recyclerView;
17     private ItemAdapter itemAdapter;
18
19     @Override
20     public View onCreateView(LayoutInflater inflater, ViewGroup
21 container, Bundle savedInstanceState) {
22
23         View view = inflater.inflate(R.layout.fragment_malat,
24 container, false);
25
26
27         recyclerView = view.findViewById(R.id.recycler_view);
28         recyclerView.setLayoutManager(new
29 LinearLayoutManager(getContext()));
30
31
32         List<String> itemList = new ArrayList<>();
33         for (int i = 1; i <= 10; i++) {
34             itemList.add("麻辣烫 " + i);
35         }
36
37         itemAdapter = new ItemAdapter(getContext(), itemList);
38         recyclerView.setAdapter(itemAdapter);
39
40         return view;
41     }
42 }

```

qingzhencaiFragment

```

01 package com.example.week7;
02 import androidx.fragment.app.Fragment;
03 import androidx.recyclerview.widget.LinearLayoutManager;
04 import androidx.recyclerview.widget.RecyclerView;
05

```

```

06 import android.os.Bundle;
07 import android.view.LayoutInflater;
08 import android.view.View;
09 import android.view.ViewGroup;
10
11 import java.util.ArrayList;
12 import java.util.List;
13
14 public class qingzhencaiFragment extends Fragment {
15
16     private RecyclerView recyclerView;
17     private ItemAdapter itemAdapter;
18
19     @Override
20     public View onCreateView(LayoutInflater inflater, ViewGroup
21 container, Bundle savedInstanceState) {
22
23         View view = inflater.inflate(R.layout.fragment_qingzhencai,
24 container, false);
25
26
27         recyclerView = view.findViewById(R.id.recycler_view);
28         recyclerView.setLayoutManager(new
29 LinearLayoutManager(getContext()));
30
31
32         List<String> itemList = new ArrayList<>();
33         for (int i = 1; i <= 10; i++) {
34             itemList.add("清真菜 " + i);
35         }
36
37         itemAdapter = new ItemAdapter(getContext(), itemList);
38         recyclerView.setAdapter(itemAdapter);
39
40         return view;
41     }
42 }

```

zhushiFragment

```

01 package com.example.week7;
02 import androidx.fragment.app.Fragment;
03 import androidx.recyclerview.widget.LinearLayoutManager;
04 import androidx.recyclerview.widget.RecyclerView;

```

```

05
06 import android.os.Bundle;
07 import android.view.LayoutInflater;
08 import android.view.View;
09 import android.view.ViewGroup;
10
11 import java.util.ArrayList;
12 import java.util.List;
13
14 public class zhushiFragment extends Fragment {
15
16     private RecyclerView recyclerView;
17     private ItemAdapter itemAdapter;
18
19     @Override
20     public View onCreateView(LayoutInflater inflater, ViewGroup
21 container,
22                             Bundle savedInstanceState) {
23
24         View view = inflater.inflate(R.layout.fragment_zhushi,
25 container, false);
26
27
28         recyclerView = view.findViewById(R.id.recycler_view);
29         recyclerView.setLayoutManager(new
30 LinearLayoutManager(getContext()));
31
32
33         List<String> itemList = new ArrayList<>();
34         for (int i = 1; i <= 10; i++) {
35             itemList.add("主食 " + i );
36         }
37
38         itemAdapter = new ItemAdapter(getContext(), itemList);
39         recyclerView.setAdapter(itemAdapter);
40
41         return view;
42     }
43 }

```

ItemAdapter

```

01 package com.example.week7;
02 import android.content.Context;
03 import android.view.LayoutInflater;

```

```

04 import android.view.View;
05 import android.view.ViewGroup;
06 import android.widget.TextView;
07
08 import androidx.annotation.NonNull;
09 import androidx.recyclerview.widget.RecyclerView;
10
11 import java.util.List;
12
13 public class ItemAdapter extends
14 RecyclerView.Adapter<ItemAdapter.ItemViewHolder> {
15
16     private List<String> dataList;
17     private Context context;
18
19     public ItemAdapter(Context context, List<String> dataList) {
20         this.context = context;
21         this.dataList = dataList;
22     }
23
24     @NonNull
25     @Override
26     public ItemViewHolder onCreateViewHolder(@NonNull ViewGroup
27 parent, int viewType) {
28         View view =
29 LayoutInflater.from(context).inflate(R.layout.itemlayout, parent,
30 false);
31         return new ItemViewHolder(view);
32     }
33
34     @Override
35     public void onBindViewHolder(@NonNull ItemViewHolder holder, int
36 position) {
37         holder.itemText.setText(dataList.get(position));
38     }
39
40     @Override
41     public int getItemCount() {
42         return dataList.size();
43     }
44
45     public static class ItemViewHolder extends RecyclerView.ViewHolder
46 {
47         TextView itemText;

```


48

```
public ItemViewHolder(@NonNull View itemView) {  
    super(itemView);  
    itemText = itemView.findViewById(R.id.item_text);  
}  
}  
}
```

colors.xml

```
01 <?xml version="1.0" encoding="utf-8"?>  
02 <resources>  
03     <color name="purple_200">#FFBB86FC</color>  
04     <color name="purple_500">#FF6200EE</color>  
05     <color name="purple_700">#FF3700B3</color>  
06     <color name="teal_200">#FF03DAC5</color>  
07     <color name="teal_700">#FF018786</color>  
08     <color name="black">#FF000000</color>  
09     <color name="white">#FFFFFFFF</color>  
10     <color name="light_blue_600">#FF039BE5</color>  
11     <color name="light_blue_900">#FF01579B</color>  
12     <color name="light_blue_A200">#FF40C4FF</color>  
13     <color name="light_blue_A400">#FF00B0FF</color>  
14     <color name="black_overlay">#66000000</color>  
15     <color name="oringin">#F3A26C</color>  
16 </resources>
```

Build.gradle

```
01 plugins {  
02     id 'com.android.library'  
03 }  
04  
05 android {  
06     namespace 'com.example.week7'  
07     compileSdk 34  
08  
09     defaultConfig {
```

```
10
11     minSdk 21
12     targetSdk 34
13     versionCode 1
14     versionName "1.0"
15
16     testInstrumentationRunner
17 "androidx.test.runner.AndroidJUnitRunner"
18     consumerProguardFiles "consumer-rules.pro"
19 }
20
21 buildTypes {
22     release {
23         minifyEnabled false
24         proguardFiles
25 getDefaultProguardFile('proguard-android-optimize.txt'),
26 'proguard-rules.pro'
27     }
28 }
29 compileOptions {
30     sourceCompatibility JavaVersion.VERSION_1_8
31     targetCompatibility JavaVersion.VERSION_1_8
32 }
33 }
34
35 dependencies {
36
37     implementation 'androidx.appcompat:appcompat:1.2.0'
38     implementation 'com.google.android.material:material:1.3.0'
39     testImplementation 'junit:junit:4.+'
40     androidTestImplementation 'androidx.test.ext:junit:1.1.2'
41     androidTestImplementation
42 'androidx.test.espresso:espresso-core:3.3.0'
43 }
```

